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L3

Solution: **B**

Taking n, R, V to be constant
Using the ideal gas equation,

$$P_i = 160 + 100 = 260 \text{ kPa}$$

$$T_i = 273.15 + 28 = 301.15 \text{ K}$$

$$T_f = 273.15 + 65 = 338.15 \text{ K}$$

$$PV = nRT$$

$$\frac{P}{T} = \frac{nR}{V} = \text{constant}$$

$$\frac{P_i}{T_i} = \frac{P_f}{T_f}$$

$$\frac{260}{301.15} = \frac{P_f}{338.15}$$

$$P_f = 291.94 \text{ kPa}$$

New reading of Pressure Gauge

$$= 291.94 - 100 = 191.9 \text{ kPa}$$

$$\% \text{age increase} = \frac{191.9 - 160}{160} \times 100\% = 20\%$$

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Solution: **D**